

ELECTRIC VEHICLE CHARGE, RANGE & MARKET ANALYSIS

Tableau Analytics & Business Intelligence Report

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1. Executive Summary

As the automotive market shifts toward electric propulsion, vehicle range, pricing, and charging infrastructure are central decision factors for both consumers and city planners. This project brings together four datasets — global EV pricing and specifications (180 models), detailed vehicle performance specifications (98 models after cleaning), Indian market EV listings (12 models), and Delhi public charging station locations (168 stations after deduplication) — into a single Tableau analytics workbook. Data cleaning was performed in MySQL Workbench; visualization, calculated fields, and dashboarding were performed in Tableau. The analysis is framed around three real-world personas defined for this project: Aarav, a city energy planner assessing charging infrastructure; Meera, an EV fleet manager assessing vehicle range and efficiency; and Ravi, a prospective buyer comparing models on price, range, and value.

2. Data Preprocessing & Methodology

Each dataset was cleaned in MySQL Workbench before being connected to Tableau. Key steps performed:

- **Text trimming:** Removed leading/trailing whitespace from key text fields (Car, Style, Transmission, VehicleType in the India dataset; Brand, Model, RapidCharge, PowerTrain, BodyStyle in the full-spec dataset; Name, Subtitle, Drive in the pricing dataset).
- **Duplicate removal:** The Delhi charging station list was reduced from 202 raw rows to 168 rows using a SELECT DISTINCT operation, stored as the charging_station_clean table.
- **Duplicate check:** The full-spec dataset was checked for duplicate Brand+Model combinations; 5 duplicate rows were removed, reducing 103 rows to 98.
- **Invalid value check:** Rows with Range_Km, TopSpeed_KmH, or PriceEuro less than or equal to zero were checked for and excluded.
- **Unit normalization (Tableau calculated fields):** Text fields containing embedded units (e.g. "335 km", "7.3 sec", "15 kW", "€42,990") were converted to numeric fields using FLOAT/REPLACE formulas, since MySQL TRIM alone does not strip embedded units.
- **LOD modeling:** A FIXED level-of-detail calculation (Underperformer Flag) was used to compare each vehicle's range against its own segment average, rather than the dataset-wide average.

Dataset Volumes After Cleaning

Dataset	Rows (raw → cleaned)	Columns
Cheapest EVs (Global)	180 → 180 (no rows removed)	11
Full EV Specifications (Global)	103 → 98	14
EV India	12 → 12 (no rows removed)	10
Delhi Charging Stations	202 → 168	8

3. Market & Vehicle Findings

Global Pricing and Range (n = 180 models)

Range across the global pricing dataset spans 95 km to 970 km, with a mean of 337 km. Listed price (Germany) spans €18,460 to €215,000, with a mean of €58,725. Drivetrain configuration is fairly evenly split: Front Wheel Drive (71 models), All Wheel Drive (64 models), and Rear Wheel Drive (45 models).

The five lowest-priced models with a listed German price are:

Model	Price (Germany, €)	Range (km)
Smart EQ fortwo coupe	18,460	100
Smart EQ forfour	19,120	95
Dacia Spring Electric	20,490	170
Smart EQ fortwo cabrio	21,720	95
Fiat 500e Hatchback 24 kWh	23,560	165

Full Vehicle Specifications (n = 98 models, post-cleaning)

Range in this dataset spans 170 km to 970 km (mean 350 km). Energy efficiency spans 104 Wh/km to 273 Wh/km (mean 190 Wh/km, where a lower figure indicates better efficiency). Every model in this particular dataset is flagged RapidCharge = Yes, so rapid-charge capability could not be used as a differentiator within this data — it reflects the composition of this dataset rather than the market as a whole.

The five most efficient models by Wh/km:

Brand	Model	Efficiency (Wh/km)	Range (km)
Lightyear	One	104	575
Hyundai	IONIQ Electric	153	250
Tesla	Model 3 Standard Range Plus	153	310
Hyundai	Kona Electric 39 kWh	154	255
Mini	Cooper SE	156	185

By drivetrain: All Wheel Drive is the most common configuration (41 models), followed by Front Wheel Drive (36) and Rear Wheel Drive (21). By vehicle segment, Segment C is the largest group (30 models), followed by Segment B (22) and Segment D (15). Average range by segment ranges from 195 km (Segment A) up to 970 km for the single Segment S model present — that single high-range vehicle should be treated as an outlier rather than representative of its segment when used for fleet planning.

Indian Market EV Listings (n = 12 models)

This is a small dataset limited to 12 electric models. Range spans 306 km to 471 km; price spans ₹13.64 lakh to ₹180 lakh. Compact SUV is the dominant body style (6 of 12 models). The Tata Nexon EV is the lowest-priced model (₹17.4 lakh, 312 km range); the Audi E-Tron GT is the highest-priced (₹180 lakh, 388

km range). With only 12 records, findings from this dataset should be treated as indicative rather than statistically representative of the full Indian EV market.

4. Charging Infrastructure Findings (Delhi, n = 168 stations)

Charging stations are unevenly distributed across the eight regional authorities present in the data. NDMC accounts for the largest share (57 stations, 34%), followed by Noida Authority (43, 26%) and SDMC (26, 15%). NKDA (16), CMRL (14), ANERT (6), NRANVP (4), and Maha Metro (2) each represent a smaller share.

Region	Stations	Share
NDMC	57	34%
Noida Authority	43	26%
SDMC	26	15%
NKDA	16	10%
CMRL	14	8%
ANERT	6	4%
NRANVP	4	2%
Maha Metro	2	1%

By charger type, three categories appear in the data: "DC-001" (103 stations, all rated 15 kW), "AC-001" (17 stations, all rated 10 kW), and "CCS/ChAdeMO/Type 2 AC" (48 stations, all rated 142 kW). It is worth flagging a labeling inconsistency here: the 142 kW stations — the highest-powered in the dataset, consistent with DC fast-charging speeds — are labeled with "AC" in their type name, while the lower-powered 15 kW stations are labeled "DC-001". This appears to be a naming inconsistency in the source data rather than a technical fact, and should be verified against the original station records before being used in planning decisions. Separately, every station in this dataset is listed as "Self Service" — no attended-service stations are present in the data.

5. Strategic Recommendations

- **For Aarav (City Energy Planner):** NDMC and Noida Authority together account for 60% of recorded charging stations. Regions with minimal presence in this dataset — Maha Metro (2 stations) and NRANVP (4 stations) — are candidates for further ground-truth investigation, since low counts here may reflect gaps in data reporting coverage as much as gaps in actual infrastructure.
- **For Meera (Fleet Manager):** Segment C and B vehicles (average range 265–330 km) make up the majority of practical fleet options and should form the baseline for range planning. The single Segment S vehicle (970 km) is a statistical outlier and should not be used to set fleet-wide range expectations.
- **For Ravi (EV Buyer):** Budget-focused buyers should consider the Dacia Spring Electric (€20,490, 170 km) or Fiat 500e (€23,560, 165 km) as the lowest-cost entry points identified in this data. Buyers prioritizing efficiency should consider the Hyundai IONIQ Electric or Tesla Model 3 Standard Range Plus, both rated at 153 Wh/km — among the best efficiency figures in the dataset.

Appendix A: Tableau Dashboards

Electric Cars Analytics Dashboard — overview page with global and India brand summary cards, BodyStyle and Brand selectors.

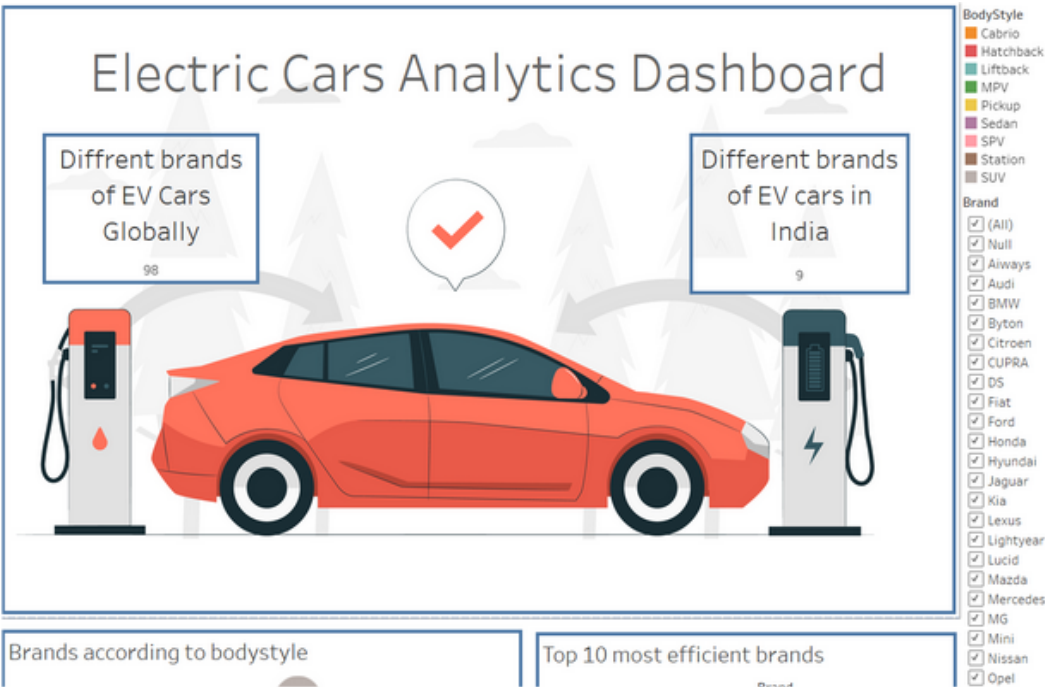


Figure 1: Electric Cars Analytics Dashboard (overview page)

Brand-by-PowerTrain breakdown and Indian EV model treemap, from the same workbook.

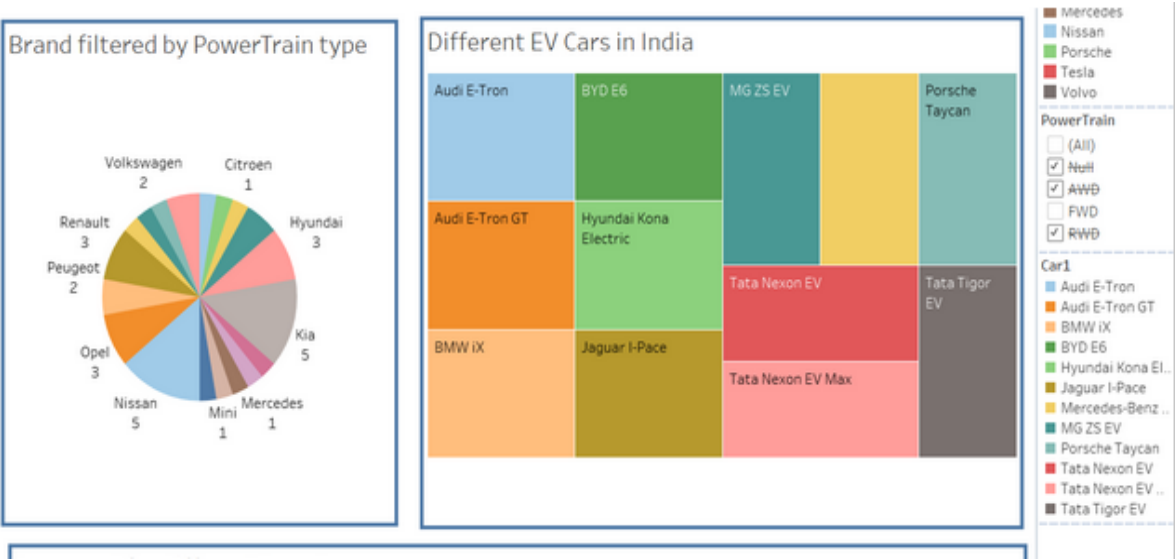


Figure 2: Brand filtered by PowerTrain type, and Different EV Cars in India

Appendix B: Data Pipeline

The diagram below reflects the actual pipeline used in this project: raw CSV files were loaded into a MySQL database (ev_analysis), cleaned via TRIM and DISTINCT operations, then connected to Tableau for calculated fields, worksheets, dashboards, and story creation. Web integration via Flask is planned but not yet implemented, and is shown accordingly.

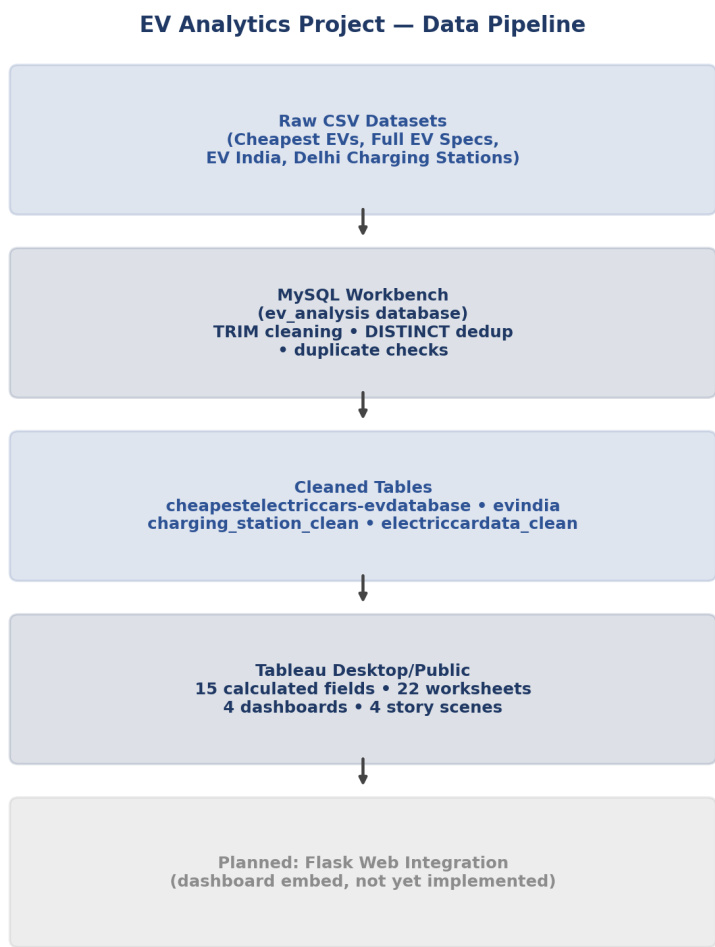


Figure 3: EV Analytics Project data pipeline